Detecting Voice Forgeries by Combining Acoustic Features with Statistical and Deep Models — Intelligent Voice Fraud Detection

Izrailov et al. (2026), in their paper *Intelligent Voice Fraud Detection Methods in Authentication Systems*, present a hybrid approach for detecting voice forgeries in voice-based biometric authentication systems. The study focuses on the vulnerability of Speaker Verification systems to impersonation attacks, in which an attacker may use a replayed recording, synthesized speech, or voice conversion in order to impersonate an authorized user. The researchers emphasize that a match between the recording's voice and the claimed identity of the speaker is not sufficient to ensure authenticity, and therefore it is necessary to incorporate Anti-Spoofing mechanisms capable of detecting features that indicate forgery. The goal of the study is to improve detection reliability by combining classical speech-signal processing methods with statistical modeling and deep learning.

The proposed method includes several processing stages. First, Preprocessing of the speech signal is performed, including removal of silent segments, loudness adjustment, conversion to a mono channel, and changing the sampling rate to a uniform frequency, such as 16kHz. Acoustic features of the Mel-Frequency Cepstral Coefficients (MFCC) and Perceptual Linear Prediction (PLP) types are then extracted. MFCC features describe aspects of the spectral envelope based on the Mel scale, while PLP combines principles of auditory perception with Linear Predictive Coding. The researchers assume that combining the two types of features may provide complementary information about the structure of the speech and about artifacts created in synthesis and voice-conversion processes.

After the feature extraction, the researchers use a Gaussian Mixture Model–Universal Background Model (GMM–UBM) to create a statistical representation of the speech recording. The background model describes features of speech from several speakers, and can be adapted to the features of a recording or speaker by means of Maximum a Posteriori Adaptation. In the experiments, 512 Gaussian components were chosen, and the distribution parameters serve as the basis for the representation passed to the classifier. At the classification stage, a relatively lightweight Convolutional Neural Network (CNN) is applied, whose purpose is to learn relations between the statistical features and to detect signs of forgery. In addition, a decision-making framework is described that includes four categories: successful verification, failed verification, strong signs of forgery, and moderate signs of forgery. However, most of the reported quantitative evaluation deals with classifying real versus fake speech, and not with separate performance for the four categories.

The researchers evaluated the method on ASVspoof 2019 Logical Access and compared it to two classical approaches, LFCC with GMM and CQCC with GMM. The combined system, MFCC+PLP+GMM+CNN, achieved an Accuracy of 97.8% and an Equal Error Rate (EER) of 4.79%, compared with an Accuracy of 94.5% and an EER of 7.44% for LFCC+GMM, and an Accuracy of 95.3% and an EER of 6.27% for CQCC+GMM. In addition, for the proposed approach a False Acceptance Rate of 4.61%, a False Rejection Rate of 4.94%, and a minDCF of 0.623 were reported. These results indicate an advantage of the hybrid system relative to the two reference methods examined, but do not make it possible to determine the separate contribution of each component in the system.

Within an additional evaluation, the researchers used data from the TED-LIUM dataset, which included 50 speakers and fake examples created by means of WaveNet and AutoVC. In this experiment, an Accuracy of about 97% and an EER of about 5.1% were reported. These results provide a certain indication of the system's ability to cope with an additional recording source, but the evaluation was limited to a small number of forgery types, and the paper does not present sufficient detail of the performance by attack type or recording conditions.

The study has several methodological limitations. First, no Ablation experiments are presented that make it possible to isolate the contribution of MFCC, PLP, GMM–UBM and the CNN, and therefore it cannot be concluded whether the improvement stems from the feature combination, from the deep classifier, or from their joint effect. Second, the implementation details of the network, the dimensions of the statistical representation, and the connection process between the components are not detailed to an extent that allows full reproduction. In addition, although the researchers present the method as suitable for practical biometric systems, no direct and detailed evaluation of inference time, memory usage, or computational cost on target hardware was carried out. The evaluation focused mainly on English speech data, and did not include a systematic examination of attacks based on Neural Audio Codecs or of robustness under varying communication conditions.

For the present study, the main contribution of the paper is in presenting a Feature Fusion approach that combines complementary acoustic features with statistical modeling and a learned classifier. Its findings provide justification for examining Baselines based on MFCC and CQCC, and for a possible extension by means of PLP or an additional statistical component. However, given a limited dataset including recordings from ASVspoof 5, from CodecFake and from M-AILABS, it should be examined whether the addition of features improves the EER and the generalization ability between the data sources, or whether it increases the risk of overfitting. Accordingly, it is recommended to start with a controlled comparison of simple features and classifiers, and only afterwards to examine the contribution of a more complex hybrid architecture. This design will make it possible to evaluate not only the improvement in performance but also its stability and the computational cost involved.

Bibliographic source:

Izrailov, K. E., Kotenko, I. V., & Chechulin, A. A. (2026). *Intelligent Voice Fraud Detection Methods in Authentication Systems*. Pattern Recognition and Image Analysis, 36(2), 391–399. https://doi.org/10.1134/S1054661826700203